required by the normative conception of the self. On that conception, they qualify as individual selves in their own rights (Braude 1995, Rovane 1998, Wilkes 1988).

Between these two extreme cases of group and multiple selves, the normative conception provides for a wide spectrum of cases, exhibiting differing degrees of rational unity within and among human beings. The conception says that whenever there is enough rational unity—though it is hard to say how much is enough—there is a rational being with its own point of view who can be rationally engaged as such and who, therefore, qualifies as an individual self in its own right.

5. *Implications for Methodological Individualism*

It is not the case that all of the facts about the Columbia Philosophy Department can be reduced to facts about its members atomistically described. Nor is it the case that all of the facts about that department can be explained by appealing to facts about its members atomistically described. Why? Because sometimes the members of the department stop reasoning from their own separate human-size points of view and begin to reason together from the point of view of the department's distinctive projects, needs, opportunities and so on. When the members of the department do this, they give over a portion of their human lives to the life of the department in a quite literal sense, so that the department itself can deliberate and act from a point of view that cannot be equated with any of their individual points of view. Outside the context of this article, this claim about the Columbia Philosophy Department might naturally be taken to contradict methodological individualism. This is because it is generally assumed that the only individuals there are or can be human individuals. That is what leads many to interpret the facts about the Columbia Philosophy Department—insofar as they cannot be properly described or explained by appealing to facts about its individual human members atomistically described—as irreducibly social phenomena. However, this interpretation fails to take account of the way in which the department instantiates the rational properties that are characteristic of the individual self (vs. a social whole).

See also: Person and Self: Philosophical Aspects; Personality and Conceptions of the Self; Self: History of the Concept; Social Identity, Psychology of

Bibliography

Bilgrami A 1998 Self-knowledge and resentment. In: Wright C et al. (eds.) *Knowing Our Own Minds*. Oxford University Press, Oxford, UK

Braude S E 1995 *First Person Plural: Multiple Personality Disorder and the Philosophy of Mind*. Rowman Littlefield, Lanham, MD

Davidson D 1984 *Inquires into Truth and Interpretation*. Oxford University Press, New York

Elster J 1979 *Ulysses and the Sirens: Studies in Rationality and Irrationality*. Cambridge University Press, New York

Evans G 1982 *Varieties of Reference*. Oxford University Press, New York

Gilbert M 1989 *On Social Facts*. Routledge, London

Korsgaard C 1989 Personal identity and the unity of agency: A Kantian response to parfit. *Philosophy and Public Affairs* **28**(2)

Levi I 1985 *Hard Choices*. Cambridge University Press, New York

Locke J 1979. In: Niddich P (ed.) *An Essay Concerning Human Understanding*. Oxford University Press, Oxford, UK

Nagel T 1970 *The Possibility of Altruism*. Clarendon, Oxford, UK

Parfit D 1984 *Reasons and Persons*. Clarendon, Oxford, UK

Perry J (ed.) 1975 *Personal Identity*. University of California Press, Berkeley, CA

Rorty A O (ed.) 1979 *The Identities of Perons*. University of California Press, Berkeley, CA

Rovane C 1998 *The Bounds of Agency: An Essay in Revisionary Metaphysics*. Princeton University Press, Princeton, NJ

Scruton R 1989 Corporate persons. *Proceedings of the Aristotelian Society* **63**

Sidgewick H 1981 *The Methods of Ethics*. Hackett, Indianapolis, IN

Wilkes K V 1988 *Real People: Personal Identity Without Thought Experiments*. Oxford University Press, New York

Wittgenstein L 1953 *Philosophical Investigations*. Macmillan, New York

C. Rovane

Self-regulated Learning

Self-regulated learning refers to how students become masters of their own learning processes. Neither a mental ability nor a performance skill, self-regulation is instead the self-directive process through which learners transform their mental abilities into task-related skills in diverse areas of functioning, such as academia, sport, music, and health. This article will define self-regulated learning and describe the intellectual context in which the construct emerged, changes in researchers' emphasis over time as well as current emphases, methodological issues related to the construct, and directions for future research.

1. *Defining Self-regulated Learning*

Self-regulated learning involves metacognitive, motivational, and behavioral processes that are personally initiated to acquire knowledge and skill, such as goal setting, planning, learning strategies, self-reinforcement, self-recording, and self-instruction. A self-regulated learning perspective shifts the focus of educational analyses from students' learning abilities and instructional environments as fixed entities to students' self-initiated processes for improving their

methods and environments for learning. This approach views learning as an activity that students do for themselves in a proactive way, rather than as a covert event that happens to them reactively as a result of teaching experiences. Self-regulated learning theory and research is not limited to asocial forms of education, such as discovery learning, self-education through reading, studying, programmed instruction, or computer assisted instruction, but can include social forms of learning such as modeling, guidance, and feedback from peers, coaches, and teachers. The key issue defining learning as self-regulated is not whether it is socially isolated but rather whether the learner displays personal initiative, perseverence, and adaptive skill in pursuing it. Most contemporary self-regulation theorists have avoided dualistic distinctions between internal and external control of learning and have envisioned self-regulation in broader, more interactive terms. Students can self-regulate their learning not only through covert cognitive means but also through overt behavioral means, such as selecting, modifying, or constructing advantageous personal environments or seeking social support. A learner's sense of self is not limited to individualized forms of learning but includes self-coordinated collective forms of learning in which personal outcomes are achieved through the actions of others, such as family members, team-mates, or friends, or through use of physical environment resources, such as tools. Thus, covert self-regulatory processes are viewed as reciprocally interdependent with behavioral, social, and environmental self-regulatory processes.

Self-regulation is defined as a variable process rather than as a personal attribute that is either present or absent. Even the most helpless learners attempt to control their functioning, but the quality and consistency (i.e., quantity) of their processes are low. Novice learners rely typically on naive forms of self-regulatory processes, such as setting nonspecific distal goals, using nonstrategic methods, inaccurate forms of self-monitoring, attributions to uncontrollable sources of causation, and defensive self-reactions. By contrast, expert learners display powerful forms of self-regulatory processes, especially during the initial phase of learning. Student efforts to self-regulate their learning has been analyzed in terms of three cyclical learning phases. Forethought phase processes anticipate efforts to learn and include self-motivational beliefs, such as self-efficacy, outcome expectations, intrinsic interest, as well as task analysis skills, such as planning, goal setting, and strategy choice. Performance phase processes seek to optimize learning efforts and include use of time management, imagery, self-verbalization, and self-observation processes. Self-reflection phase processes follow efforts to learn and provide understanding of the personal implication of outcomes. They include self-judgment processes, such as self-evaluation and attributions, and self-reactive processes, such as self-satisfaction and adaptive/defensive inferences. Because novice learners fail to use effective forethought processes proactively, such as proximal goal setting and powerful learning strategies, they must rely on reactive processes occurring after learning attempts that often have been unsuccessful. Such unfortunate experiences will trigger negative self-evaluations, self-dissatisfaction, and defensive self-reflections—all of which undermine self-motivation necessary to continue cyclical efforts to learn. By understanding self-regulation in this cyclical interactive way, qualitative as well as quantitative differences in process can be identified for intervention.

2. *Intellectual Context for Self-regulated Learning Research*

Interest in students' self-regulated learning as a formal topic emerged during the 1970s and early 1980s out of general efforts to study human self-control. Promising investigations of children's use of self-regulatory processes like goal setting, self-reinforcement, self-recording, and self-instruction, in such areas of personal control as eating and task completion prompted educational researchers and reformers to consider their use by students during academic learning. Interest in self-regulation of learning was also stimulated by awareness of the limitations of prior efforts to improve achievement that stressed the role of mental ability, social environmental background of students, or qualitative standards of schools. Each of these reform movements viewed students as playing primarily a reactive rather than a proactive role in their own development. In contrast to prior reformers who focused on how educators should adapt instruction to students based on their mental ability, sociocultural background, or achievement of educational standards, self-regulation theorists focused on how students could proactively initiate or substantially supplement experiences designed to educate themselves.

Interest in self-regulation of learning emerged from many theoretical sources during the 1970s and 1980s. For example, operant researchers adapted the principles and technology of B. F. Skinner for personal use, especially the use of environmental control, self-recording, and self-reinforcement. Their preference for the use of single-subject research paradigms and time series data was especially useful for individuals seeking greater self-regulation of learning. During this same time period, phenomenological researchers shifted from monolithic, global views of how self-perceptions influenced learning to hierarchical, domain-specific views and began developing new self-concept tests to assess functioning in specific academic domains. The research of Hattie Marsh, Shavelson, and others was especially influential in gaining new currency for the role of academic domain self-concepts in learning. During this era, social learning researchers, such as Bandura, Schunk, and Zimmerman,

shifted their emphasis from modeling to self-regulation and renamed the approach as social cognitive. They identified a new task-specific motive for learning—self-efficacy belief, and they linked it empirically to other social cognitive processes, such as goal setting, self-observation, self-judgment, and self-reaction. During this era, researchers such as Corno, Gollwitzer, Heckhausen, Kuhls, and others, resurrected volitional notions of self-regulation to explain human efforts to pursue courses of learning in the face of competing events. In their view, self-regulatory control of action can be undermined by ruminating, extrinsic focusing, and vacillating—which interfere with the formation and implementation of an intention.

Also during the 1970s and 1980s, suppressed writings of Vygotsky were published in English that explained more fully how children's inner speech emerges from social interactions and serves as a source of self-control. Cognitive behaviorists, such as Meichenbaum, developed models of internalization training on the basis of Vygotsky's description of how overt speech becomes self-directive. During this same era, cognitive constructivists shifted their interest from cognitive stages to metacognition and the use of learning strategies to explain self-regulated efforts to learn. The research and theory of Flavell played a major role in effecting this transition by describing self-regulation in terms of metacognitive knowledge, self-monitoring, and control of learning.

Research on self-regulation was also influenced by the emergence of goal theories during the 1970s and 1980s. Locke and Lathan showed that setting specific, proximal, challenging but attainable goals greatly influenced the effectiveness of learners' efforts to learn. Theorists such as Ames, Dweck, Maehr, Midgley, and Nicholls identified individual differences in goal orientations that affected students' efforts to learn on their own. These researchers found that learning or mastery goal orientations facilitated persistence and effort during self-directed efforts to learn, whereas performance or ego goal orientations curtailed motivation and achievement. During this same period, another perspective emerged that focused on the role of intrinsic interest in learning. Deci, Harackiewicz, Lepper, Ryan, Vallerand, and others demonstrated that perceptions of personal control, competence, or interest in a task were predictive of intrinsic efforts to learn on one's own. This focus on intrinsic motivation was accompanied by a resurgence of research on the role of various forms of interest in self-directed learning by a host of scholars in Europe as well as elsewhere, such as Eccles, Hidi, Krapp, Renninger, Schiefele, Wigfield, and others. During these same decades, research on self-regulation of learning was influenced by cybernetic conceptions of how information is processed. Researchers such as Carver, Scheier, and Winne demonstrated the important role of executive processes, such as goal setting and self-monitoring, and feedback control loops in self-directed efforts to learn.

3. Changes in Emphasis Over Time

Before the 1980s, researchers focused on the impact of separate self-regulatory processes, such as goal setting, self-efficacy, self-instruction, volition, strategy learning, and self-management with little consideration for their broader implications regarding student learning of academic subject matter. Interest in the latter topic began to coalesce in the mid-1980s with the publication of journal articles describing various types of self-regulated learning processes, good learning strategy users, self-efficacious learners, and metacognitive engagement, among other topics (Jan Simons and Beukhof 1987, Zimmerman 1986). By 1991, a variety of theories and some initial research on self-regulated learning and academic achievement was published in special journal articles and edited textbooks (Maehr and Pintrich 1991, Zimmerman and Schunk 1989). These accounts of academic learning, which described motivational and self-reactive as well as the metacognitive aspects of self-regulation, spurred considerable research. By the mid-1990s, a number of edited texts had been published chronicling the results of this first wave of descriptive research and experimental studies of self-regulated learning (Pintrich 1995, Schunk and Zimmerman 1994). The success of these empirical studies stimulated interest in systematic interventions to students' self-regulated learning and the results of these implementations emerged in journal articles and textbooks by the end of the 1990s (Schunk and Zimmerman 1998).

4. Emphases in Current Theory and Research

There is much current interest in understanding the self-motivational aspects of self-regulation as well as the metacognitive aspects (Heckhausen and Dweck 1998). Self-regulated learners are distinguished by their personal initiative and associated motivational characteristics, such as higher self-efficacy beliefs, learning goal orientations, favorable self-attributions, and intrinsic motivation, as well as by their strategic and self-monitoring competence. The issue of self-motivation is of both practical as well as theoretical importance. On the practical side, researchers often confront apathy or helplessness when they seek to improve students' use of self-regulatory processes, and they need viable methods for overcoming this lack of motivation. On the theoretical side, researchers need to understand how motivational beliefs interact with learning processes in a way that enhances students' initiative and perseverance. A number of models have included motivational and learning features as interactive components, such as Pintrich's self-schema model, Boerkaert's three-layered model, Kuhl's action/state control model, and Bandura, Schunk, and Zimmerman's cyclical phase model. These models are designed to transcend conceptual barriers between

learning and motivational processes and to understand their reciprocal interaction. For example, causal attributions are not only expected to affect students' persistence and emotional reactions but also adaptations in their methods of learning. Each of these models seeks to explain how learning can become self-motivating and can sustain effort over obstacles and time.

A second issue of current interest is the acquisition of self-regulatory skills in deficient populations of learners. Study skill training programs have been instituted with diverse populations and age groups of students. These intervention programs have involved a variety of formats, such as separate strategy or study skill training classes, strategy training classes linked explicitly to subject matter classes, such as in mathematics or writing, or personal trainer services in classrooms and counseling centers. This research is uncovering a fascinating body of evidence regarding the metacognitive, motivational, and behavioral limitations of at-risk students. For example, naive learners often overestimate their knowledge and skill, which can lead to understudying, nonstrategic approaches, procrastination, and faulty attributions. Successful academic functioning requires accurate self-perceptions before appropriate goals are set and strategies are chosen. Many of these interventions are predicated on models that envision self-regulatory processes as cyclically interdependent, and a goal of these approaches is to explain how self-fulfilling cycles of learning can be initiated and sustained by students.

5. *Methodological Issues*

Initial efforts to measure self-regulated learning processes relied on inventories in which students are asked to rate their use of specific learning strategies, various types of academic beliefs and attitudes, typical methods of study, as well as their efforts to plan and manage their study time. Another method is the use of structured interviews that involved open-ended questions about problematic learning contexts, such as writing a theme, completing math assignments, and motivating oneself to study under difficult circumstances. The latter form of assessment requires students to create their own answers, and experimenters to train coders to recognize and classify various qualitative forms of self-regulatory strategies. Although both of these approaches have reported substantial correlations with measures of academic success, they are limited by their retrospective or prospective nature—their focus on the usual or preferred methods of learning. As such, they depend on recall or anticipatory knowledge rather than on actual functioning under taxing circumstances. To avoid these limitations, there have been efforts to study self-regulation on-line at various key points during learning episodes using speak-aloud, experimenter questioning, and various performance and outcome measures. Experimental studies employing the latter measures offer the best opportunity to test the causal linkage among various self-regulatory processes, but, even this approach has potential shortcomings, such as inadvertent cuing or interference with self-regulatory processes.

Two common research design issues have emerged in academic interventions with at-risk populations of students: the lack of a suitable control group, and the reactive effects of self-regulatory assessments. Educators have asked self-regulation researchers to provide assistance to students who are often in jeopardy of expulsion, and it is unethical to withhold treatment from some of these students for experimental purposes. One solution is to use intensive within-subject, time-series designs in which treatments are introduced in sequential phases. However, as students begin to collect and graph data on themselves, they become more self-observant and self-reflective—which can produce unstable baselines. This, in turn, can confound causal inferences about the effectiveness of other self-regulatory components of the intervention.

6. *Future Directions for Research and Theory*

6.1 *Role of Technology*

The computer has been recommended as an ideal instrument to study and enhance students' self-regulation. Program menus can be faded when they are no longer needed and performance processes and outcomes can be logged in either a hidden or overt fashion. Computers provide the ultimate feedback to the experimenter or the learner because results can be analyzed and graphed in countless ways to uncover underlying deficiencies. Winne is developing a specially designed computer learning environment designed to study and facilitate self-regulation by the student.

6.2 *Developmental Research*

Relatively little attention has been devoted to forms of self-regulation that can be performed by young children. Very young children have difficulty observing and judging their own functioning, are not particularly strategic in their approach to learning, and tend to reason intuitively from beliefs rather than evidence. Their self-judgments of competence do not match their teachers' judgments until approximately the fifth grade. However, there is reason to expect that simple forms of self-regulatory processes begin to emerge during the early years in elementary school. For example, there is evidence that children can make self-comparisons with their earlier performance at the outset of elementary school. These issues are con-

nected to the key underlying issue of how self-regulation of learning develops in naturalistic as well as designed instructional contexts.

6.3 Out-of-school Influences

There is increasing research showing that non-academic factors such as peer groups, families, and part-time employment strongly affect school achievement. Schunk and Zimmerman have discussed the forms of social influence that these social groups can have on students' self-regulatory development. Brody and colleagues have found that parental monitoring of their children's activities and standard setting regarding their children's performance were very predictive of the children's academic as well as behavior self-regulation, which in turn was predictive of their cognitive and social development. Martinez-Pons has recently reported that parental modeling and support for their children's self-regulation was predictive of the youngsters' success in school. More attention is needed about the psychosocial origins of self-regulatory competence in academic learning.

6.4 Role of Teachers

The recent focus on academic intervention research has uncovered evidence that teachers often conduct classrooms where it is difficult for students to self-regulate effectively. For example, teachers who fail to set specific instructional goals, are ambiguous or inconsistent about their criteria for judging classroom performance, give ambiguous feedback about schoolwork, make it difficult for students to take charge of their learning. Of course, students who enter such classes with well-honed self-regulatory skills possess personal resources that poorly self-regulated learners do not possess. Such self-regulatory experts can turn to extra-classroom sources of information, can deduce subtle unspecified criteria for success, and can rely on self-efficacious beliefs derived from earlier successful learning experiences.

Regarding the development of self-regulated learners, few teachers ask students to make systematic self-judgments about their schoolwork, and as a result, students are not prompted or encouraged to use self-regulatory subprocesses such as self-observation, self-judgment, and self-reactions. Students who lack awareness of their functioning have little reason to try to alter their personal methods of learning. Finally, teachers who run classrooms where students' have little personal choice over their goals, methods, and outcomes of learning can undermine students' perceptions of control and assumption of responsibility for their classroom outcomes.

See also: Learning to Learn; Motivation, Learning, and Instruction; Self-efficacy: Educational Aspects

Bibliography

Heckhausen J, Dweck C S (eds.) 1998 *Motivation and Self-regulation Across the Lifespan*. Cambridge University Press, Cambridge, UK

Jan Simons P R, Beukhof R (eds.) 1987 *Regulation of Learning*. SVO, Den Haag, The Netherlands

Maehr M L, Pintrich P R (eds.) 1991 *Advances in Motivation and Achievement: Goals and Self-regulatory Processes*. JAI Press, Greenwich, CT, Vol. 7

Pintrich P (ed.) 1995 *New Directions in College Teaching and Learning: Understanding Self-regulated Learning*. Jossey-Bass, San Francisco, No. 63, Fall

Schunk D H, Zimmerman B J (eds.) 1994 *Self-regulation of Learning and Performance: Issues and Educational Applications*. Erlbaum, Hillsdale, NJ

Schunk D H, Zimmerman B J (eds.) 1998 *Self-regulated Learning: From Teaching to Self-reflective Practice*. Guilford Press, New York

Zimmerman B J (ed.) 1986 Self-regulated learning. *Contemporary Educational Psychology* **11**, Special Issue

Zimmerman B J, Schunk D H (eds.) 1989 *Self-regulated Learning and Academic Achievement: Theory, Research, and Practice*. Springer, New York

B. J. Zimmerman

Copyright © 2001 Elsevier Science Ltd.
All rights reserved.

Self-regulation in Adulthood

Self-regulation is one of the principal functions of the human self, and it consists of processes by which the self manages its own states and actions so as to pursue goals, conform to ideals and other standards, and maintain or achieve desired inner states. Many experts use self-regulation interchangeably with the everyday term self-control, although some invoke subtle distinctions such as restricting self-control to refer to resisting temptation and stifling impulses.

1. Scope of Self-regulation

Most knowledge about self-regulation can be grouped into four broad and one narrower category. The most familiar is undoubtedly impulse control, which refers to regulating one's behavior so as not to carry out motivated acts that could have harmful or undesirable consequences. Dieting, responsible sexual behavior, recovery from addiction, and control of aggression all fall in the category of impulse control.

A second category is affect regulation, or the control of emotions. The most common form of this is the attempt to bring oneself out of a bad mood or bring an end to emotional distress. In principle, however, affect regulation can refer to any attempt to alter any mood or emotion, including all attempts to induce, end, or prolong either positive or negative emotions. Affect regulation is widely regarded as the most difficult and problematic of the major spheres, because most people